# Analytical Chemistry Cumulative Examination
## April 26, 2002

1. Electrospray ionization mass spectrometry is an exceptionally useful technique for analysis of high mass chemicals. Please answer each of the following questions about this technique.

a) Prior to introduction of this method, quadrupole mass analyzers were generally limited to analysis of molecules with molecular weights of 2,000 Daltons or lower. ESI-MS has greatly changed this situation, particularly for analysis of proteins. Please explain why this is so and how ESI makes this possible. What are current mass limits for protein analysis? (10 points)

b) Data processing of electrospray mass spectra can have a number of complications. Please explain what those are and what methods (both computational and experimental) can be used to overcome those problems. (10 points)

c) There are various models for the electrospray ionization process. Two of these are Dole's charged residue model and Iribarne and Thomson's ion extraction ("Coulomb explosion") model. Explain, with the help of a suitable, clearly labeled, diagram the basic idea of the Iribarne and Thomson model for the mechanism of electrospray. (15 points)

2. In an early paper devoted to electrospray Fourier transform mass spectrometry McLafferty and his students (*J. Amer. Soc. Mass Spectrom.*, **1993**, 4 , 557-565) covered some of the advantages of this method. Please answer the following questions. (a) What is the principle advantage of using the FTMS over a quadrupole MS for this application? (b) What order of magnitude resolving power can be obtained for protein molecular ions in using FTMS? (c) Charge assignment can be a difficult problem in electrospray MS. How can FTMS be used to advantage for such assigments? (15 points).

3. The April , 2002 issue of *Analytical Chemistry* has an "Accelerated Article" by Egelhofer and coworkers which describes a new strategy for identifying proteins by MALDI-TOF-MS. Briefly summarize the approach described in the paper and explain the advantages and disadvantages of the method. What types of application would this approach be best for and for what applications would it be inappropriate? How is sample introduced and manipulated in this method? What kinds of samples were used to demonstrate operation of the new approach? How were results verified? (50 points)

3. UV-Vis in Titrations (30 points total)

Draw the titration curve (relative absorbance vs. volume of titrant) that corresponds to the following scenarios a, b, and c below. Show your work. No credit will be given for a graph without the supporting data and explanation. Assume that the total volume is not significantly changed during the titration (so that you don't have to account for dilution effects.)

Here are the relevant conditions:

- The reaction of interest is: $A + T \rightarrow P$
- Your analyte is "A" and is the only species that absorbs visible radiation in the starting sample solution. "A" does not absorb at 700 nm, but does absorb at 400 nm.
- You titrate your sample solution with the titrant 'T", which absorbs at 700 nm, but does not absorb at 400 nm.
- A product is formed, "P", from the reaction of "A" with "T". "P" has the same absorption characteristics as "A", in that it does not absorb at 700 nm, but does absorb at 400 nm. In addition, it has the same extinction coefficient as "A".
- When one molecule of "A" reacts with one molecule of "T", the reaction is essentially complete and irreversible…a good condition of a titration.

(a) (10 points) Draw the titration curve of absorbance at 400 nm as a function of titrant added. (Also, indicate the endpoint on the titration curve.)

(b) (10 points) Draw the titration curve of absorbance at 700 nm as a function of titrant added. (Also, indicate the endpoint on the titration curve.)

(c) (10 points) What are the advantages of titrations over direct UV-Visible absorbance spectroscopy of a sample of unknown analyte concentration?